

January 31st, 2020

Via GCKey

Mr. Claude Doucet
Secretary General
Canadian Radio-television and
Telecommunications Commission
Ottawa, Ontario
K1A 0N2

Dear Mr. Doucet:

Re: Request to add conditions of licence for several television stations to multiplex some of its stations (2019-1119-9, 2019-1124-8, 2019-1123-0 and 2019-1121-4)

Corus is pleased to provide its responses to the request for information contained in the Commission's letter dated January 23rd, 2020 regarding the above-mentioned applications. We have reproduced the questions below, followed by our responses in bold.

Technical

- 1) Please comment on the equality of video of the OTA transmission, in terms of image resolution for each station's signal, before and after multiplexing of the signal.

As a preliminary comment, Corus suggests that the equality of the video in terms of image resolution for each station's signal before and after multiplexing of the signal to the home viewer, is highly subjective. The proposed changes will not be perceivable by most OTA viewers. We provide larger context to this comment below.

The CRTC's glossary provides the following criteria when defining high definition television:

High definition television refers to a set of standards for television that offer the highest resolution and sharpest image possible. High definition usually refers to 720 vertical lines of video format resolution or more (for example, 720p, 1080i, 1080p). Standard definition refers to 480 vertical lines of video format (480i, 480p). In these examples, the letter "p" designates "progressive," which means that each scan (vertical) line is progressively scanned in each frame; that is, the entire picture is scanned into each frame. The letter "i" designates



"interlaced," which means that a television frame is updated in parts over two slices of time (the first slice draws the odd numbered scan lines, and then the next draws the even number scan lines). (Sources: CRTC Staff; sonyhdivinfo.com; and Wikipedia: "High-definition television," 8 April 2009)

Corus broadcasts OTA signals according to the Advanced Television Systems Committee (ATSC) standard A/53, which provides for variation in video image formats.

Currently, Corus OTA transmission conforms to the SMPTE 274M video standard, with 1920x1080i resolution (1080i). After multiplexing, OTA transmission will conform to the SMPTE 296 video standard, with 1280x720p resolution (720p). Both 1080i and 720p are High Definition (HD) video signals.

720p HD transmission is commonplace in the USA where multiplex is widely deployed. The American Broadcasting Company (ABC) and Fox Broadcasting Company (Fox) have extensively deployed 720p in multiplexed ATSC 1.0 transmitters, such as WABC and WNYW in New York and KABC and KTTV in Los Angeles.

The perceived difference between 1080i and 720p video image formats depends on the size and quality of the receiving television and the consumers viewing angle and distance from the television. While the 720p image size is smaller than 1080i, the perceived sharpness of an interlace image is less than a progressive image. As a result, a "difference" between 1080i and 720p to the home viewer is highly subjective.

The planned change will affect the bitrate of the delivered signal, from 19.39 Mbps currently broadcast for single program service to approximately 9.7 Mbps per program service (and approximately 6.5 Mbps per program service at CHEX) in the proposed multiplex. This reduction in bitrate is achieved by increasing the level of compression in the broadcast station's encoder/multiplexer. ATSC 1.0 stipulates the use of MPEG-2 compression. MPEG-2 is "lossy compression" meaning a decrease in the bitrate may be noticed by a discerning viewer. The discernable difference is affected, however, by the quality of the encoder/multiplexer utilized.

Prior to proceeding with the application Corus conducted a series of tests at the proposed bitrates. The conclusion of our engineering staff was the proposed changes will not be perceivable by most OTA viewers due to the genre of content typically broadcast on the channels (sports programming is most affected by compression) subject to the application and the quality of the apparatus planned to achieve the multiplex.



The aspect ratio for both the current OTA transmission and proposed multiplex transmission is 16:9.

- 2) Given that ISED currently assigns call signs based on the transmitter and that a transmitter can only have one call sign associated with it, please comment on how the OTA transmissions of the multiple stations will be identified after the signals are multiplexed and broadcast off a single transmitter (i.e. What call signs will be used to identify each station? How will each station be displayed on a TV for a user who is scanning for it after the transition?).

Each station will continue to be displayed on a TV in the same manner with the same call signs as they are today. Viewers will not notice any difference in the identification of each station as they scan for any one of the stations.

Following discussions with ISED regarding the applications before the Commission, Corus has submitted applications with ISED to amend each of the Broadcasting Certificates for those stations in which the transmitter will multiplex an additional channel. The amended certificates will allow those stations that are being multiplexed to maintain their current call signs. We have included as Appendix 1 – 6 copies of the applications submitted to ISED.

While ISED has advised that they cannot amend and issue the revised broadcast certificates until the CRTC has issued a decision on the Part 1 Applications to multiplex, they have issued authorization numbers to each of the stations which Corus plans to multiplex. Table 1 below provides the call letters and the ISED Authorization numbers.

Table 1

Channel	Station/Channel Identification	Call Letters Pre Multiplex	Call Letters Post Multiplex	ISED Authorization #
27.1	CHBC-DT Kelowna	CHBC-DT	CHBC-DT	010299682-001
27.2	CHKL-DT Kelowna	CHKL-DT	CHKL-DT	010299695-001
32.1	CHBC-DT-1 Penticton	CHBC-DT-1	CHBC-DT-1	010299574-001
32.2	CHKL-DT-1 Penticton	CHKL-DT-1	CHKL-DT-1	010299569-001
20.1	CHBC-DT-2 Vernon	CHBC-DT-2	CHBC-DT-2	010299563-001
20.2	CHKL-DT-2 Vernon	CHKL-DT-2	CHKL-DT-2	010299556-001
13.1	CKWS-DT Kingston	CKWS-DT	CKWS-DT	010299647-001



13.2	CIII-DT-2 Bancroft	CIII-DT-2	CIII -DT-2	010299638-001
14.1	CIII-DT-6 Ottawa	CIII-DT-6	CIII-DT-6	010299925-001
14.2	CKWS-DT-2 Prescott	CKWS-DT-2	CKWS-DT-2	010790904-001
12.1	CHEX-DT	CHEX -DT	CHEX-DT	010299650-001
12.2	CIII-DT-27	CIII-DT-27	CIII-DT-27	010300622-001
12.3	CKWS-DT-1	CKWS-DT-1	CKWS-DT-1	010300562-001

- 3) In the Supplementary Briefs, Corus stated that “Broadcasting Regulatory Policy CRTC 2015-24 states that when electing to shut down transmitters, broadcasters may lose certain regulatory privileges, including mandatory distribution on the basic service, and the right to request simultaneous substitution. It is important to note that Corus is not proposing to shut down its transmitters noted in the application above as it did with 44 rebroadcasting transmitters in October 2018. Rather, Corus is proposing to change the transmission point of the signal through the process of multiplexing. Corus’ regulatory privileges in the effected region should therefore be unaffected.” In some cases, the region served by the transmitter being shut down is different from the region of the new transmission site. Please clarify, for each of your applications, what is meant by the “effected region”.

In the applications, Corus referred to the fact that while transmitters are being shutdown, the signals that those transmitters provided will continue to be available through an innovative process of multiplexing from another transmitter. The term ‘effected region’ referred to the area served by the transmitter both pre and post multiplexing.

For ease of explanation, we have attached Appendices 7–10 which illustrate the impact of the ‘effected region’ utilizing the Noise-Limited Bounding Contour (NBLC). And while the NBLC provides a more accurate picture than the DTV Urban Contour (DUC) by including those viewers who employ an antenna to receive the OTA signal, it is important to take into account that this is a worst case scenario. It should be taken into consideration that both the DUC and the NBLC relate specifically to those viewers who receive the services over-the-air and does not take into account that those viewers who receive the signal via cable/satellite and will not be affected.

Further, as we discussed in our applications , through the earlier process of shutting down transmitters, Corus’ experience has been that for those viewers who contacted us concerned that they had lost the Global signal, we were able to help the viewer re-establish the Global signal by repositioning their antenna and/or rescanning their television in the majority of cases.

- 4) The following tables reflect the amendments that have been proposed. In order to provide more complete information regarding the impact of the proposed amendments on existing population served, please complete the following tables.

Information regarding the DTV Urban Contour (DUC) population has been populated below in Table 2. However, we believe that this does not provide the full picture of existing population served. While DUC refers to those viewers who can receive OTA transmissions with no antennae, it does not include DTV Noise-Limited Bounding Contour (NLBC) which includes those viewers who can receive the OTA signal with a mid-level quality indoor antennae.¹ Corus has therefore included the NLBC numbers below to include a more accurate frame of reference.

Having said that, as noted in question 3, it should be taken into consideration that both the DUC and the NLBC relate specifically to those viewers who receive the services over-the-air and does not take into account those viewers who receive the signal via cable/satellite and will not be affected.

Further, as we discussed in our applications, through the earlier process of shutting down transmitters, Corus' experience has been that for those viewers who contacted us concerned that they had lost the Global signal, we were able to help the viewer re-establish the Global signal by repositioning their antenna and/or rescanning their television in the majority of cases.

Table 2

Applications	Signal to be multiplexed (existing)	Multiplexing transmitter (new)	Existing DTV Urban Contour (DUC) population that will remain in new DUC	Existing DUC no longer receiving service	Existing DTV Noise-Limited Bounding Contour (NLBC) population that will remain in new NLBC	Existing NLBC no longer receiving service
2019-1124-8	CIII-DT-2	CKWS-DT	0	14,597	90,642	173,460
2019-1123-0	CKWS-DT-1	CHEX-DT	8,551	35,941	173,264	700
	CIII-DT-27	CHEX-DT	184,075	72,752	997,546	11,294
2019-1119-9	CHBC-DT	CHKL-DT	321,208	0	369,861	500
	CHBC-DT-1	CHKL-DT-1	118,425	0	267,625	0
	CHBC-DT-2	CHKL-DT-2	93,839	0	298,635	0

¹ Mid-level quality indoor antennae are readily available at most big-box stores for less than \$60.

2019-1121-4	CKWS-DT-2	CIII-DT-6	0	7,733	48,541	2,959
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Programming

- 5) In application 2019-1119-9, you are requesting that CHKL DT Kelowna (rebroadcaster of CHAN DT- Vancouver) multiplex its digital television signal that would allow CHBC-DT Kelowna (mother station) to transmit from this site. Please confirm that you will continue to abide by your programming conditions of licence for CHBC-DT, including those related to locally reflective programming, in the event that your application is approved. Please also confirm that your locally reflective programming will continue to be focused on Kelowna.

As identified in table 1 included in our response to question 2, each distinctive station will be maintained. Corus confirms that this means we will continue to abide by our programming conditions of licence for CHBC-DT Kelowna, including those related to locally reflective programming, in the event that our application is approved. We further confirm that the locally reflective programming for CHBC-DT will continue to be focused on Kelowna.

- 6) In Policy framework for local and community television², the Commission imposed regulatory measures to ensure that Canadians continue to have access to local programming that reflects their needs and interest.

The Commission determined that news programming must be locally reflective and must meet all of the following criteria:

- the subject matter relates specifically to the market a station is licensed to serve;
 - it portrays an onscreen image of the market by, for example, including its residents or officials or featuring coverage of its municipal or provincial government; and
 - it is produced by the station's staff or by independent producers specifically for the station.
- a) In application 2019-1124-8, it is proposed that CIII-TV-2 be multiplexed from the site of CKWS-DT, and the provided maps show that the population within CIII-DT-2's existing local service area as enclosed by the DUC would fall completely outside the DUC of CKWS-DT. Please confirm that you will continue to abide by your programming conditions of licence for CIII-DT-2, including those related to locally reflective programming, in the event that your application is approved. Please indicate how this change will affect the nature of your locally reflective programming broadcast by CIII-DT Toronto.



² Broadcasting regulatory Policy CRTC 2016-224

As identified in table 1 included in our response to question 2, each distinctive station will be maintained. Corus confirms that this means we will continue to abide by our programming conditions of licence for CIII-DT-2. CIII-DT-2 is currently a rebroadcast channel of CIII-DT-41 and therefore does not have any programming that differs from CIII-DT-41, consequently there will be no change to the nature of the locally reflective programming broadcast by CIII-DT Toronto.

- b) In application 2019-1123-0, it is proposed that CKWS-DT-1 Brighton and CIII-DT-27 Peterborough will be multiplexed from the site of CHEX DT Peterborough. The provided maps show that a portion of the population within the local service area enclosed by the DUC of CKWS-DT-1 and CIII-DT-27 would fall outside the DUC of CIII-DT-41. Please confirm that you will continue to abide by your respective conditions of licence in programming for CIII-DT-27 and CKWS-DT-1, including those related to locally reflective programming, in the event that your application is approved. Please indicate how this change will affect the nature of your locally reflective programming broadcast by CIII-DT Toronto, and that the programming on CKWS-DT-1 will be focused on Brighton while the programming for CHEX-DT will be focused on Peterborough for CHEX-DT.

As identified in table 1 included in our response to question 2, each distinctive station will be maintained. Corus confirms that this means we will continue to abide by your respective conditions of licence in programming for CIII-DT-27 and CKWS-DT-1. CIII-DT-27 is currently a rebroadcast channel of CIII-DT-41 and therefore does not have any programming that differs from CIII-DT-41, consequently there will be no change to the nature of the locally reflective programming broadcast by CIII-DT Toronto. CKWS-DT-1 is a rebroadcast channel of CKWS-DT and therefore does not have any programming that differs from CKWS-DT. While CKWS-DT-1 does have a COL that permits exception on commercial availabilities on the Brighton station separately from those broadcast on CKSW-DT for each original station-produced programming it broadcasts exclusively on its Brighton station each week³, Corus is not required to and does not currently take advantage of that opportunity. Corus confirms that the locally reflective programming for CHEX-DT will continue to be focused on Peterborough.

- c) In application 2019-1121-4, it is proposed that CKWS-DT-2 be multiplexed from the site of CIII-DT-6, and provided maps show that the population within CKWS-DT-2's existing local service area as enclosed by the DUC would fall completely outside the DUC of CIII-DT-6. Please confirm that you will continue to abide by your programming conditions of licence for CKWS-DT-2 Prescott, including those related to locally reflective programming, in the

³ Broadcasting Decision CRTC 2017-150 paragraph 32.

event that your application is approved. Please also indicate how this change will affect the nature of your locally reflective programming broadcast by CKWS-DT Kingston.

As identified in table 1 included in our response to question 2, each distinctive station will be maintained. Corus confirms that this means we will continue to abide by our programming conditions of licence for CKWS-DT-2. CKWS-DT-2 is currently a rebroadcast channel of CKWS-DT and therefore does not have any programming that differs from CKWS-DT, consequently there will be no change to the nature of the locally reflective programming broadcast by CKWS-DT Kingston.

Distribution

- 7) In The regulatory framework for the distribution of digital television signals, Broadcasting Public Notice CRTC 2003-61, the Commission established that “determinations concerning whether and under what terms a licensed multicast service might be authorized for distribution by a BDU would be reached by the Commission, on a case-by-case basis, at the time it considers the licence application for the multicast service” (see paragraph 81). In light of the above-noted case-by-case approach, please clearly indicate the terms under which your proposed multiplexed station should be distributed by Canadian broadcasting distribution undertakings (BDUs)?

We interpret Broadcasting Public Notice CRTC 2003-61 as requiring a case-by-case approach in relation to new services, not services that are already authorized for distribution.

To be clear, in these applications, Corus does not seek broadcasting licences for new multiplexed television services. We seek to amend the conditions of licence of television stations, which are already authorized for distribution. More specifically, we have requested new conditions of licence that would permit the multiplex of existing transmitters in order to allow other existing Corus-owned stations to transmit from a shared site.

Broadcasting Public Notice CRTC 2003-61 states: “the Commission would be disposed to authorize the distribution of new and innovative multicast services in preference to those that merely duplicate other services.”⁴

First, the amendments we propose would represent an “innovative” solution to address significant financial and operational challenges. Continued erosion of conventional television audiences, and significant new costs associated with the mandatory repack of the 600 MHz band have increased financial pressures on these stations. The amended conditions would enable

⁴ See paragraph 81.

us to maintain the existing authorized rebroadcast signals in affected communities while mitigating costs.

Second, granting these amendments which will ensure continued distribution of these stations in their communities would plainly be in the public interest.

Finally, we note our amendments would result in minimal distribution for customers in the affected areas.

Corus trusts that the information provided is satisfactory. We would be please to provide any further information at the Commission's request.

Thank you for your attention to this matter.

Sincerely,



Karen Gifford
Senior Director, Regulatory Compliance and Licensing
Law Department
Corus Entertainment Inc.

c.c. Manon Auger, CRTC

Attachments

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